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PI Name:	Masson, Patrick H. Ph.D.		
Project Title:	Using Brachypodium distachyon to Study the Molecular Mechanisms That Underlie Directional Root-Growth Responses to Low-Speed 2-Dimensional Clinorotation		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
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Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
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Task Description:	As key components of bioregenerative life-support systems, plants will be exposed to multiple stresses associated with spaceflight. Experiments carried out in Space have dramatically improved our understanding of plant responses to microgravity. In turn, this novel knowledge has allowed the design of better culturing methods that enable plant development and reproduction in Space. Yet, to design even better growth strategies and develop optimized genotypes favorable to plant growth and productivity during spaceflight, we need to better understand the molecular machinery that contribute to the perception of, and response to gravity and/or microgravity. Unfortunately, the large cost associated with Space experimentation and limited accessibility to the resources needed for such experimentation, limit our ability to acquire this new knowledge. Consequently, utilization of devices that mimic microgravity is viewed as a viable alternative. The 2D-clinostat has often been utilized for this purpose. By constantly changing the orientation of biological specimens relative to gravity, these devices prevent gravity-vector detection, thereby triggering behavioral responses that are similar to those expressed under true microgravity. The 2D-clinostat has often been used to estimate the sensitivity of plant organs to different doses of gravi-stimulation. In this case, low doses of gravi-stimulation are provided at the beginning of the experiment. Stimulated seedlings are subsequently clinorotated for 4-5 hours, allowing for a curvature to develop in response to the initial stimulus. Yet, the data derived from these analyses are likely to be significant only if the clinorotation does not introduce artifacts of its own. In experiments involving Arabidopsis thaliana seedlings, we previously observed a strong clinostat effect on root skewing on hard agar surfaces. Follow-up experiments involving Brachypodium distachyon confirmed a clear impact of low-speed clinorotation on directional root-tip bending. These experiments also suggested an important role for the first cycle of clinorotation in setting up a specific direction of gravistimulation as long as it is provided for a sufficiently long time period. The general objective of this research project will be to investigate the effect of low speed clinorotation on Brachypodium root growth. Specific objectives include: Aim 1: Develop high throughput time-lapse imaging to describe the development of B. distachyon root curvature responses to clinorotation at high spatio-temporal resolution. Aim 2: Use transgenic B. distachyon seedlings that express auxin-level, auxin-function, and auxin-efflux carrier reporters to investigate the effect of clinorotation on auxin distribution and signaling along the root tip, and on the distribution of auxin transporters between flanks of clinorotated root tips. Aim3: Use a combination of Genome Wide Association Study (GWAS) and RNA Seq strategies to identify candidate genes that may contribute to root-tip response to clinorotation, and use reverse genetics to investigate the role(s) of such candidates and of gravitropism-related genes in the process. Overall, this project will improve our understanding of how low-speed clinorotation affects root growth behavior. While our observations suggest a need for greater caution in using a low-speed 2D clinostat as a microgravity-mimic system, they also suggest a novel method to investigate the molecular mechanisms that contribute to gravity sensing in plants, one of the main objectives of the Space Biology program. This project will introduce undergraduate students to the idiosyncrasies of scientific research in Space Biology, while also putting a strong emphasis on undergraduate and K-12 education and public outreach.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will improve our understanding of how low-speed clinorotation affects root growth behavior, while re-emphasizing a need for greater caution in using a low-speed 2D clinostat as a microgravity-mimic system. Our research is likely to result in the development of a novel method to investigate the molecular mechanisms that contribute to gravity sensing in plants, one of the main objectives of the Space Biology program. In the longer term, the basic information acquired during these studies should enable the design of plants that are better suited to function as key components of bioregenerative life support systems for long-term space exploration missions, either during travel under microgravity, or exposed to distinct gravity environments on exotic planets. They may also allow the development of novel crop cultivars with increased productivity in natural and/or agricultural ecosystems on Earth. With this project, we are able to introduce undergraduate students to the idiosyncrasies of scientific research in Space Biology, in addition to developing effective educational materials to introduce K-12 and undergraduate students, as well as the general public, to the difficult concepts of gravity perception by plants and its importance for agricultural production on Earth and during spaceflight.
	When Arabidopsis thaliana or Brachypodium distachyon seedlings growing on the surface of agar-based media are positioned on a clinostat rotating at low speed along an axis perpendicular to the growth surface, their roots curve in a direction that is dictated by the polarity of rotation 1. The direction of tip curvature correlates with the direction of gravistimulation imposed on the root tip during a phase of the clinorotation cycle that is long enough to trigger a response. For instance, the roots of wild type Col seedlings positioned vertically in the clinostat at the beginning of clinorotation will curve in a direction that is dictated by the gravistimulus provided during the first ½ turn of clinorotation (from vertical down to vertical up). However, if the clinorotation is initiated with wild type seedlings positioned horizontally on the clinostat, the direction of curvature will not be dictated by the direction of gravistimulation during the initial 1/4 turn (15 sec) of clinorotation bringing the root tips from a horizontal to a vertical orientation. Instead, the direction of root curvature correlates with the direction of gravistimulation during the following ½ turn of clinorotation that rotates the root tip from vertical-down to vertical-up. These results suggest the clinostat effect on root tip orientation is a consequence of gravistimulation during clinorotation, where the polarity of gravistimulation during the initial phase of rotation dictates the overall direction of tip curvature provided it is long enough to set this response polarity. Subsequent rounds of clinorotation simply allow that curvature to proceed in the pre-set direction. To test this hypothesis, we investigated the effect of low speed clinorotation on root-tip growth of Arabidopsis thaliana wild type and gravity-deficient mutant seedlings, including those with defects in the PHOSPHOGLUCOMUTASE (PGM) gene (which encodes an enzyme necessary for starch synthesis and statolith sedimentation in the statocytes 2), in TOC132 (which encodes a protein that contributes to protein import into the plastids and interacts with ARG1 to orchestrate gravisensing 3) and in the LAZY 2, 3, and 4 genes (which also contribute to early phases of gravity signal transduction in the statocytes 4,5). In last year's report, we showed that these mutations affect the roots' ability to curve in response to clinorotation, supporting a role for gravity signal transduction in modulating the clinostat effect in this plant. This year, we continued our investigations of the connection between gravity sensing and the clinostat effect on root tip curvature using Arabidopsis thaliana mutant seedlings, initiated experiments aimed at testing the use of clinorotation at

Task Progress:	different speeds to evaluate gravisensitivity in plants, and tested representative accessions of Brachypodium distachyon, a model plant for the monocots (including most cultivated cereal crops), for differential sensitivities to the clinostat effect. Our results suggest a possible clinostat-effect-based method to evaluate plant gravisensitivity, and demonstrate large variability in root growth responses to clinorotation between Brachypodium distachyon accessions. The results obtained this year are very encouraging, providing the foundation for a new, simple approach to evaluate root gravisensitivity in plants. However, Covid-19-related distancing guidelines imposed significant restraints on our ability to efficiently complete these experiments. Therefore, we will be requesting a 1-year no-cost extension to complete this work and use genome-wide association studies in Brachypodium distachyon to identify loci that contribute to root -tip response to clinorotation. Comparing wild type and gravitropism-deficient Arabidopsis thaliana mutants for root tip responses to clinorotation. To investigate a possible connection between gravity sensing and clinosensitivity in Arabidopsis thaliana roots, we continued investigations of root tip curvature responses to different speeds of clinorotation using both wild type Col and gravitropism-deficient toc132-1 and pgm-1 mutant seedlings. Results indicated significant differences in mutant root tip responses to clinorotation relative to wild type. Mathematical modeling of root curvature responses to clinorotation. The classical method to evaluate plant-organ gravi-sensitivity involves gravistimulating the plants for short time periods, then clinorotating them along an axis parallel to the seedlings for a few hours to allow formation of a curvature response to the initial stimulus. This leads to different levels of curvature depending on the initial period of stimulation. Mathematical models can then be fit to the data, allowing evaluation of gravisensitivity. Utilization of a logarithmic function to model the data allows estimation of the presentation time, which is defined as the minimal time of gravistimulation needed to allow the plant organ to initiate a productive curvature. The presentation time is often viewed by researchers as a good estimate of root gravisensitivity even though a hyperbolic model has also been suggested as a powerful alternative by the Perbal laboratory. In last year's progress report, we established presentation times of 2.4 min for Col and 6.4 min for toc132-2, compatible with a decrease in root sensitivity to gravistimulation in toc132-2 mutant relative to wild type. This year, we investigated the possibility of using a similar approach to evaluate Arabidopsis root sensitivity to clinorotation. In this experiment, we clinorotated wild type Col, toc132-1, and toc132-2 mutant seedlings at different speeds (0.25, 0.5, 1, and 2 rpm), then quantified the root curvature angles after a period of 24 h in darkness. A logarithmic function was fit to the data, allowing calculation of a new parameter we named Realization Time, defined as the minimal time needed during the initial phase of clinorotation to fix the polarity of root-tip curvature response and develop a significant curvature. By analogy with the presentation time, the realization time can be estimated by fitting a logarithmic model to root curvature angles for different periods of 1/2 turns during clinorotation (1/2 inverse of clinorotation speed), then extending the model to its intersection with the X-axis. Interestingly, a preliminary experiment testing this approach allowed estimation of realization times of 29 sec for wild type Col, 44 sec for mutant toc132-1, and 48 sec for toc132-2. Hence, the toc132 mutant seedlings appear less sensitive to clinorotation relative to wild type, again mimicking their defect in gravisensitivity. As previously discussed in this report, we suggest the clinostat effect on root curvature might be a simple manifestation of a gravitropic response to the directional exposure to gravity experienced by roots during the first 1/2 turn of clinorotation. If this exposure is longer than the realization time, the root will respond by curving in this direction. Hence, the realization time may be a useful alternative estimate of root gravisensitivity. Brachypodium distachyon accessions display distinct root curvature responses to low-speed clinorotation. As initially mentioned in our grant proposal, the roots of Brachypodium distachyon also display strong root tip responses to clinorotation. This year, we investigated whether different accessions display distinct abilities to respond to low-speed clinorotation by developing distinct root tip curvatures in the direction of clinorotation. Therefore, we tested three Brachypodium accessions chosen for their distinct root growth responses to gravi-stimulation, for the ability of their roots to curve in response to clinorotation at different speeds. The results clearly showed different impacts of low-speed clinorotation on root-tip curvature between accessions, with Gaz8 displaying strong responses at 0.2, 0.5, and 0.7 rpm, Bd21-3 showing a weak response at 0.2 rpm, and Ron2 displaying no curvature at any of the clinorotation conditions tested. We are currently in the process of evaluating the possibility to estimate realization times for each one of these accessions as well as for the ~100 additional accessions currently available in the lab, to ultimately develop a Genome-Wide Association Study (GWAS) to identify loci that contribute to the variation in this trait. Conclusion. The social distancing guidelines imposed by Covid-19-pandemic restrictions have significantly affected our ability to effectively conduct wet-lab research on this project last year. However, we were still able to evaluate possible connections between root curvature responses to clinorotation and gravitropism, develop a clinostat-based method to evaluate gravi- and/or clino-sensitivity, and demonstrate altered abilities of Arabidopsis gravitropism-defective mutant roots to develop tip curvatures upon clinorotation relative to wild type. We also demonstrated large differences between Brachypodium distachyon seedlings in their roots' abilities to curve in response to low-speed clinorotation, setting the ground for new genome-wide association study (GWAS) investigations of this behavior and possible identification of contributing loci. We hope to request a 1-year no-cost extension to this project to allow completion of these studies and final testing of the connection between root responses to clinorotation and gravisensitivity. These studies will improve our understanding of the way plants in general, and roots in particular, respond to the constraints imposed by clinorotation. Considering our observations of root tip curvatures with hallmarks of graviresponses under low-speed clinorotation, this work also calls for increased caution in the use of clinostats to mimic microgravity exposure in plant biology.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Abstracts for Journals and Proceedings	Moen AD, Su S-H, Groskopf RM, Baldwin K, Vesperman B, Masson PH. "Using Low-Speed 2-Dimensional Clinorotation to Study Gravi-sensitivity in Arabidopsis and Brachypodium." Presented at the 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020

Abstracts for Journals and Proceedings	Su S-H, Stephens SD, Campbell A-M, Levine H, Masson PH. "Using Brachypodium distachyon to Investigate Monocot Plant Adaptation to Spaceflight." Presented at the 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020
Articles in Peer-reviewed Journals	Su S-H, Keith MA, Masson PH. "Gravity signaling in flowering plant roots." Plants. 2020 Oct;9(10):1290. Review. https://doi.org/10.3390/plants9101290 ; PMID: 33003550; PMCID: PMC7601833 , Oct-2020